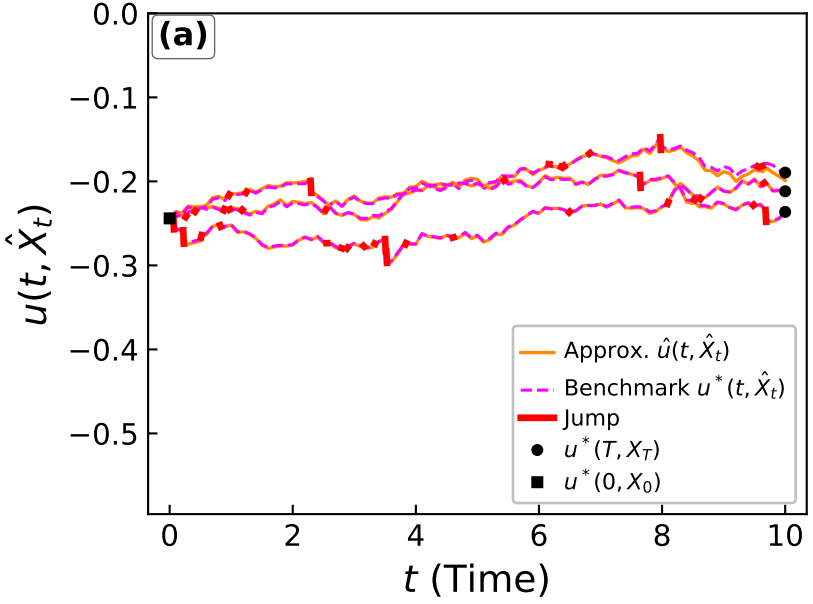
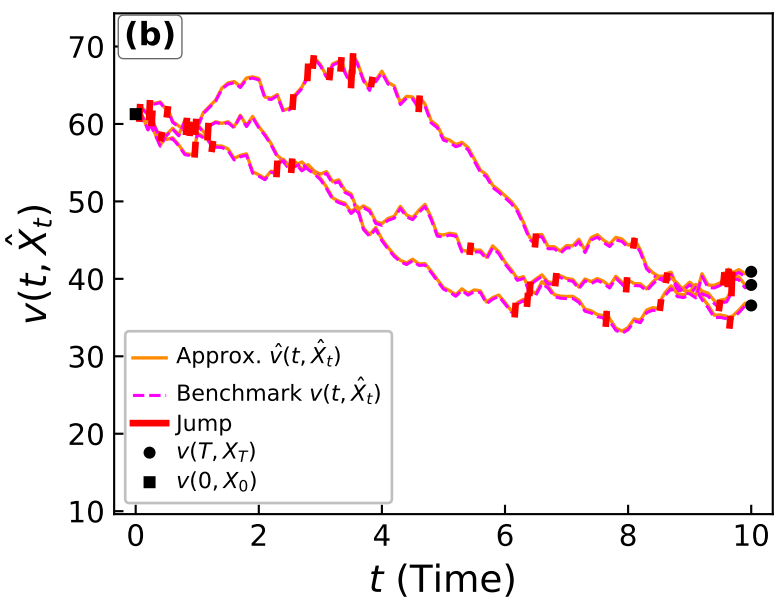
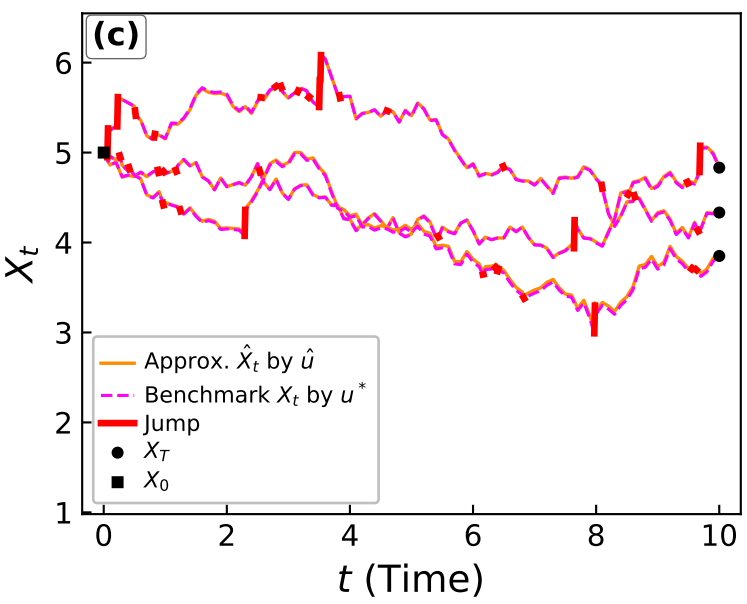

(a)
$u(t, \hat{X}_t)$
$t$ (Time)
Approx. $\hat{u}(t, \hat{X}_t)$
Benchmark $u^*(t, \hat{X}_t)$
Jump
$u^*(T, X_T)$
$u^*(0, X_0)$


(b)
$v(t, \hat{X}_t)$
$t$ (Time)
Approx. $\hat{v}(t, \hat{X}_t)$
Benchmark $v(t, \hat{X}_t)$
Jump
$v(T, X_T)$
$v(0, X_0)$


(c)
$X_t$
$t$ (Time)
Approx. $\hat{X}_t$ by $\hat{u}$
Benchmark $X_t$ by $u^*$
Jump
$X_T$
$X_0$